CSA04 – OPERATING SYSTEMS
TASK 1 – EVALUATE THE REQUIREMENTS OF EACH COMPUTING ENVIRONMENT

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1. Aim

To evaluate the operating-system requirements of different computing environments in a smart hospital based on response time, reliability, throughput, battery consumption and security, and to justify the most suitable design.

2. Ranked Requirements of Each Environment

The hospital contains different devices and servers. Their requirements are ranked according to their importance to the device's main function.

Computing environment	Ranked requirements (highest to lowest)	Justification
Bedside patient monitors	1. Response time 2. Reliability 3. Security 4. Throughput 5. Battery	Must detect abnormal readings and generate alarms within 10 milliseconds. Response time and reliability are critical to patient safety.
Nurses' tablets	1. Battery 2. Security 3. Reliability 4. Response time 5. Throughput	Must operate throughout a shift, securely access patient records and remain portable.
Doctors' desktops	1. Security 2. Reliability 3. Response time 4. Throughput 5. Battery	Must provide secure access to patient information and support multiple applications.
Central database server	1. Reliability 2. Security 3. Throughput 4. Response time 5. Battery	Patient records must remain available, accurate and protected while serving many users.
MRI imaging server	1. Throughput 2. Reliability 3. Response time 4. Security 5. Battery	Must process large MRI and CT images efficiently using multiple processors.
Partner hospital network	1. Security 2. Reliability 3. Throughput 4. Response time 5. Battery	Must securely and reliably share patient records among three hospitals.

3. Conflicting Requirements

1. Fast response and high throughput may conflict when many processes compete for CPU time.
2. Strong security checks may introduce additional processing delays.
3. High performance and long battery life may conflict on nurses' tablets.
4. Continuous database availability may conflict with maintenance and software updates.

4. Requirements Critical to Patient Safety

- Patient monitors must generate alarms within the required 10-millisecond deadline.
- Patient records must be accurate and continuously available to authorized staff.
- Emergency monitoring should continue locally even if the network or internet connection fails.
- Patient information must be protected from unauthorized access, alteration and loss.

5. Application of Operating-System History

OS development	**Application in the smart hospital**
Batch systems	Processing medical reports and historical records in groups.
Time-sharing systems	Allowing multiple doctors and staff members to use shared systems.
Personal computer systems	Supporting doctors' desktop workstations.
Mobile operating systems	Supporting nurses' handheld tablets and portable applications.
Real-time operating systems	Supporting bedside patient monitors that require predictable alarm response.

6. Alternative Concept and Complexity Analysis

Alternative: Use one general-purpose operating system, such as a standard Linux distribution, on every hospital device instead of selecting an operating system according to each environment.

Factor	**Different OS approach**	**Single general-purpose OS**
Alarm response	A properly configured and validated RTOS can provide a guaranteed 10 ms deadline.	A standard general-purpose configuration does not guarantee a 10 ms worst-case deadline; scheduling delay may be unpredictable.
Memory	A lightweight OS can reduce memory use on small monitoring devices.	A full general-purpose OS may need more memory and storage on small devices.
Administration	Different platforms require separate updates, skills and	Platform standardization can simplify some administration, but

	maintenance.	device-specific configuration and support are still needed.
Battery	A mobile OS can use power-management features designed for tablets.	A general-purpose OS may be less optimized for the tablet's battery and hardware.
Reliability	The OS can be selected to match each device's safety and availability needs.	One OS may not meet the real-time, low-power and high-availability needs of every device.

Complexity Analysis

Memory requirement: It depends on the OS build, enabled services and device drivers. A lightweight RTOS generally uses less memory than a full general-purpose OS. Big-O notation alone cannot give the actual memory size; it describes how resource use grows with input size.

Administration effort: Managing different OS platforms has higher platform diversity and training effort. A single OS can simplify standardization, although hardware-specific support and configuration are still required.

Worst-case alarm delay: A validated real-time design must demonstrate that its worst-case response time is no more than 10 ms. A normal general-purpose OS does not provide that guarantee, so its worst-case scheduling delay may be unbounded relative to the required deadline.

Big-O note: The alarm scheduling guarantee is a timing property, not a Big-O complexity result. For a simple alarm check over one sensor reading, processing is O(1) per reading; storing n readings requires O(n) space if all readings are retained. The same basic algorithm can be used on either OS, but only a suitable real-time system can provide a verified deadline.

7. Evaluation

Using one general-purpose operating system everywhere may reduce platform diversity, but it cannot efficiently satisfy all hospital requirements. Different operating systems provide better real-time performance, battery efficiency, reliability and processing capabilities. Therefore, a mixed operating-system design is more suitable.

8. Result

Thus, the requirements of all six hospital computing environments were evaluated. A combination of real-time, mobile, desktop, clustered and distributed computing systems is recommended for a safe, reliable and efficient smart hospital.